

Machine Learning Project

Loan Approval Prediction

Using Machine Learning to Automate Loan Decisions



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4,269

Total Records



99.9%

Model Accuracy



62%

Approval Rate

📋 Loan Approval Prediction Challenge

📌 Primary Goal

- Predict whether a loan application will be **Approved** or **Rejected**
- Automate the decision-making process using machine learning
- Reduce manual review time and improve efficiency

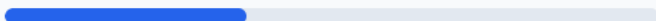
💡 Why This Matters

- Faster, more consistent lending decisions
- Reduces risk of bad loans and defaults
- Improves customer experience

📊 Current Challenge Statistics

37%

Rejection Rate



4,269

Total Applications



62%

Approval Rate



 Loan Approval Dataset Analysis

 Dataset Details

 File
loan_approval_dataset.csv

 Size
4,269 rows

No Missing Values

 Feature Distribution

12
Total Features

10
Numeric

3
Categorical

 Key Features

no_of_dependents
Numeric
0-5

education
Categorical
Graduate/Not

self-employed
Categorical
Yes/No

income_annum
Numeric
200K-9.9M

loan_amount
Numeric
300K-39.5M

loan_term
Numeric
2-20 months

cibil_score
Numeric
300-900

asset_values
Numeric
4 types